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Verbrauchte Zeit	59 Minuten 50 Sekunden
Bewertung	46,87 von 50,00 (93,73%)

Frage 1

Vollständig

Nicht bewertet

General information:

- Not counting this first "task" (i.e. the statutory declaration below), there are 12 problems with 50 points in total.
- Read all problems and possible answers carefully and take your time to think about them.
- Multiple-choice questions: Choosing wrong answers leads to deduction of points within that question. However, you cannot get less than 0 points on a question.
- Numerical questions: Wrong answers do not lead to point deductions.
- For numerical answers, use "." for the decimal point. In case you get an error message or the question is marked as not answered, try using "," instead.
- When it says that you can use a pocket calculator, you may also use the calculator of your operating system.
- After some questions, there is an optional text field. This field will not influence the automatic grading. Here, you can write down subresults of your calculations. It may be used to your advantage at the end of the semester. There will be no other form of submitting information, i.e., no emails or Moodle uploads with photos or pdfs after the exam.
- You can ask questions during the exam only via the chat window and only directly to the exam supervisor. Your multiple-choice answers may be displayed in random order. So, don't refer to them by their label.
- You have 60 minutes for the exam. After this time is expired, your exam will automatically be submitted. There is no overtime.
- Once you have submitted your exam, please state so in the chat window, e.g. write "submitted", don't wait for a reply, and then just leave the Zoom meeting.
- Strictly forbidden: All forms of audio or video recording by the candidates during the examination. This includes, but is not limited to, screenshots, downloads and any copy of the exam content (digital or analogue). Also after the exam, you are not allowed to copy or share or pass on the exam material in any way whatsoever.
- The Zoom session of the exam may be recorded. (By law, recordings must not be kept longer than 72 hours. We will delete it even earlier in case everything runs smoothly, which is what we expect.)

Statutory declaration:

With clicking on "I agree" below, you agree to this statutory declaration ("Eidesstattliche Erklärung"): "I hereby confirm that I complete this closed-book exam on my own and only using the allowed materials, in particular without any written or electronic notes and without any help from or communication with other people. I understand that violation of this declaration may cause legal consequences."

Wählen Sie eine Antwort:

- ☒ I agree
- ☐ I do not agree

Frage 2

Richtig

Erreichte Punkte 4,00 von 4,00

Which statements about the generalization error and the empirical risk are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. The generalization error is the expected loss on future data for a given model. ✓
- ☐ b. Estimating the risk via samples from a test set, which have not been used in training, avoids the problem of underfitting.
- ☐ c. The generalization error has the form $\int \int_{\mathcal{X} \times \mathcal{Y}} L(y, g(\vec{x}; \vec{w})) p(\vec{x} | y) dy d\vec{x}$.
- ☒ d. The empirical risk on a dataset with n samples has the form $R_{\text{emp}} = \frac{1}{n} \sum_{i=1}^n L(y_i, g(\vec{x}_i; \vec{w}))$. ✓
- ☒ e. A common assumption for estimation of the generalization error via the empirical risk is that data are independent and identically distributed (i.i.d.). ✓
- ☐ f. A good estimation of the generalization error via the empirical risk generally requires use of a differentiable loss function.
- ☒ g. A good estimation of the generalization error via the empirical risk generally requires a large number of samples. ✓
- ☐ h. A good estimation of the generalization error via the empirical risk generally requires that data points are uniformly distributed.

Frage 3

Teilweise richtig

Erreichte Punkte 1,87 von 4,00

Which statements about supervised machine learning are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. ROC curves allow to evaluate a classifier independently of class distribution and independently of misclassification cost. They plot TPR versus TNR. ✗
- ☒ b. Cross validation can help to optimize model hyperparameters. ✓
- ☐ c. Having a small training error but a large test error indicates the problem of underfitting.
- ☒ d. Model variance typically increases with increasing model class complexity. ✓
- ☐ e. The Bayes Optimal Classifier is a probabilistic model that makes the most probable prediction for a new sample. It requires use of a loss (or cost) function.
- ☐ f. Bias typically increases with increasing model class complexity.
- ☐ g. The unavoidable error in the bias-variance decomposition cannot be decreased (i.e. cannot be beaten) by the model training, but it can be increased by training a model poorly.
- ☐ h. Supervised machine learning uses explicit knowledge to design models deductively.

Frage 4

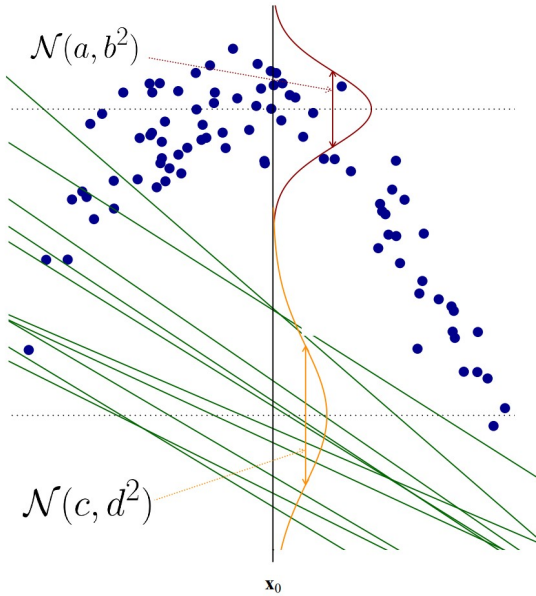
Richtig

Erreichte Punkte 4,00 von 4,00

Bias variance decomposition for quadratic loss:

Similar to what you saw in the lecture, consider the following data points and model distributions.

The vertical axis is the y-axis (corresponding to the labels), the horizontal axis the x-axis of a single feature. The green lines are different trained models. The blue dots are sampled datapoints. $\mathcal{N}(\mu, \sigma^2)$ denotes the 1-d normal distribution with mean μ and variance σ^2 . The brown and orange distributions depict the underlying distributions of labels (brown) and model predictions (orange), respectively, at a given input x_0 .



We also use the usual notation convention from the lecture, i.e.:

Let Z_I be a sample set of I elements and $g(\mathbf{x}_0; \mathbf{w}(Z_I))$ denote a trained model evaluated at $\mathbf{x}_0$, where the model parameters $\mathbf{w}$ are dependent on the respective training set Z_I .

Set the variables in the plot as follows:

$$a = 2$$

$$b = 4$$

$$c = -3$$

$$d = 6$$

Compute the values of the following terms:

- What is the model variance at x_0 ? Answer: ✓
- What is the unavoidable error at x_0 ? Answer: ✓
- What is $\mathbb{E}_{y \mid \mathbf{x}_0} (y - \mathbb{E}_{Z_I} (g(\mathbf{x}_0; \mathbf{w}(Z_I))))^2$? Answer: ✓
- What is the expected prediction error at x_0 ? Answer: ✓

Frage 5

Vollständig

Nicht bewertet

This text field is optional and will not influence the automatic grading. Here, you can write down subresults or your thoughts on the specific tasks. It may be used to your advantage at the end of the semester.

model variance: $\text{Var}(N(c, d^2)) = d^2 = 6^2 = 36$

unavoidable error: $\text{Var}(y|x_0) = b^2 = 4^2 = 16$

bias: $E[N(a, b^2)] - E[N(c, d^2)] = a - c = 2 - (-3) = 5$

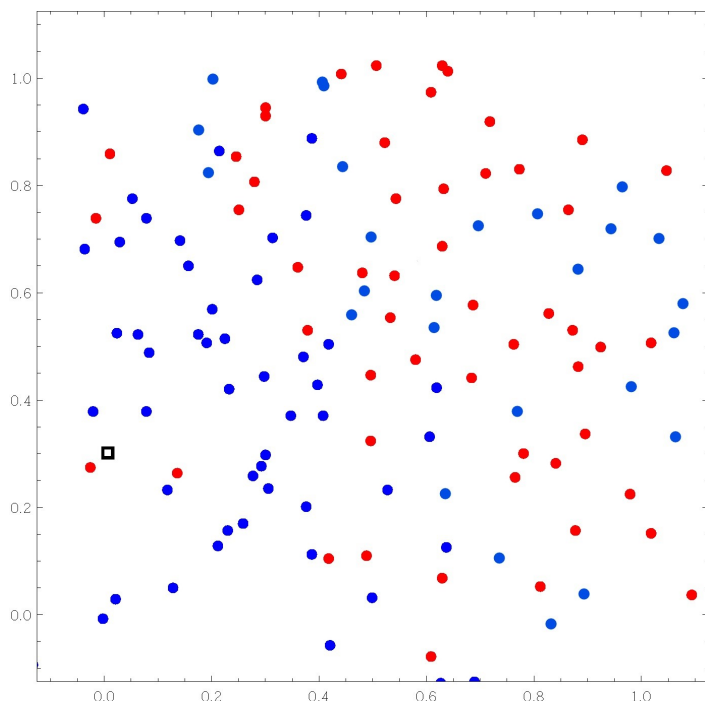
$\text{EPE} = \text{model variance} + \text{unavoidable error} + \text{bias}^2 = 36 + 16 + 25 = 77$

Frage 6

Richtig

Erreichte Punkte 4,00 von 4,00

k -nearest neighbors (k -NN): Consider the figure below. There are n training data points in total and more blue points ($y = -1$) than red points ($y = +1$). A test data point $\mathbf{x}_0$ is shown by a black square, and its label is denoted as y_0 .



Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. For $k=1$: $y_0 = -1$.
- ☒ b. For $k=3$: $y_0 = -1$ ✓
- ☒ c. For $k=n$: $y_0 = -1$ ✓
- ☐ d. $k=1$ leads to underfitting.
- ☐ e. $k=n$ corresponds to large model complexity.
- ☒ f. k -NN can also be used for regression. ✓
- ☒ g. k -NN requires use of a distance measure for measuring closeness. ✓
- ☐ h. k -NN can work with labeled and unlabeled data.

Frage 7

Richtig

Erreichte Punkte 5,00 von 5,00

Assume that currently 1% of the Austrian population have Covid, i.e. they are "positive".

Assume that a certain Covid antigen test has 95% sensitivity and 90% specificity.

Now a person comes to you and says: "I just made this Covid antigen test. And it showed a positive result, i.e. it showed that I am ill."

What is the probability that this person in fact is healthy, i.e. does not have Covid?

Remark: You do not need to know the Austrian population size. Give it a variable name (or any number you like). It will drop out of your calculation. The result does not depend on the population size.

You can use a pocket calculator. Give the answer as a real number between 0 and 1, rounded to two decimal places. Do not give the answer in %. Use "." for the decimal point. Number format example: 0.67

Antwort: 0,91

**Frage 8**

Vollständig

Nicht bewertet

This text field is optional and will not influence the automatic grading. Here, you can write down subresults or your thoughts on the specific tasks. It may be used to your advantage at the end of the semester as mentioned in the exam info file.

$$P=0.01$$

$$N=1-P=0.99$$

$$\frac{TP}{P}=0.95 \implies TP=0.95P=0.95 \cdot 0.01=0.0095$$

$$\frac{TN}{N}=0.9 \implies TN=0.9N=0.9 \cdot 0.99$$

$$FP=N-TN=0.99-(0.9 \cdot 0.99)=0.099$$

$$\frac{FP}{PP}=\frac{FP}{TP+FP}=0.9124$$

Frage 9

Richtig

Erreichte Punkte 5,00 von 5,00

Consider the following classification results of a binary classifier tested on 8 samples:

$\hat{y}$	y
0.1	1
-0.8	-1
1.0	-1
0.5	-1
1.9	1
-0.1	-1
1.4	1
-0.5	-1

Compute the Area Under Curve (AUC) of the ROC curve.

You can use a pocket calculator. Give the answer as a real number between 0 and 1, rounded to two decimal places. Do not give the answer in %. Use "." for the decimal point. Number format example: 0.67

Antwort: 0,87

**Frage 10**

Vollständig

Nicht bewertet

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g

1,9 +1

1,4 +1

1,0 -1

0,5 -1

0,1 +1

-0,1 -1

-0,5 -1

-0,8 -1

$$AOC = \frac{13}{15} = 0.87$$

AOC**curve****xxx**

xxxxx

xxxxx

Frage 11

Richtig

Erreichte Punkte 4,00 von 4,00

Which statements about support vector machines (SVMs) are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. One of the KKT conditions has the form: $\alpha_i h_i(\text{w}) \geq 0$ for all $i = 1, \dots, l$.
- ☐ b. In the dual problem, the dual function $\mathcal{L}$ is maximized with respect to $\mathbf{w}$ and $\mathbf{b}$.
- ☒ c. Support vector regression uses the magnitude (and not only the sign) of the discriminant function. ✓
- ☒ d. A sample $\mathbf{x}_i$ is a support vector if the corresponding Lagrange parameter α_i is strictly larger than 0. ✓
- ☐ e. Kernels allow to transform data into a lower-dimensional space where separability can be easier achieved.
- ☒ f. The mathematical problem (constrained convex optimization) has a global solution. ✓
- ☒ g. The Gaussian kernel has the form: $k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{x}_i - \mathbf{x}_j\|^2\right)$. ✓
- ☐ h. SVMs work effectively if the number of dimensions (features) d is much smaller than the number of samples l .

Frage 12

Richtig

Erreichte Punkte 4,00 von 4,00

Find the solution $\mathbf{w}_1^*, \mathbf{w}_2^*$ to the following constrained optimization problem with respect to $\mathbf{w}_1, \mathbf{w}_2$:

$$\begin{aligned} &\text{minimize} \quad \frac{1}{2}(\mathbf{w}_1^2 + \mathbf{w}_2^2) \\ &\text{subject to} \quad 10 - \mathbf{w}_1 - 3\mathbf{w}_2 \leq 0 \end{aligned}$$

Hint and help: The task has been designed such that the solutions are integers.

- $\mathbf{w}_1^* = 1$ ✓
- $\mathbf{w}_2^* = 3$ ✓

Frage 13

Vollständig

Nicht bewertet

This text field is optional and will not influence the automatic grading. Here, you can write down subresults or your thoughts on the specific tasks. It may be used to your advantage at the end of the semester as mentioned in the exam info file.

$$L = \frac{1}{2}(\mathbf{w}_1^2 + \mathbf{w}_2^2) + \alpha(10 - \mathbf{w}_1 - 3\mathbf{w}_2)$$

$$\frac{\partial L}{\partial \mathbf{w}_1} = \mathbf{w}_1 - \alpha = 0 \implies \mathbf{w}_1^* = \alpha$$

$$\frac{\partial L}{\partial \mathbf{w}_2} = \mathbf{w}_2 - 3\alpha = 0 \implies \mathbf{w}_2^* = 3\alpha$$

$$\frac{\partial L}{\partial \alpha} = 10 - \mathbf{w}_1 - 3\mathbf{w}_2 = 0 \implies 10 - \alpha - 3(3\alpha) = 0 \implies 10 - 10\alpha = 0 \implies \alpha = 1$$

$$\mathbf{w}_1^* = \alpha = 1$$

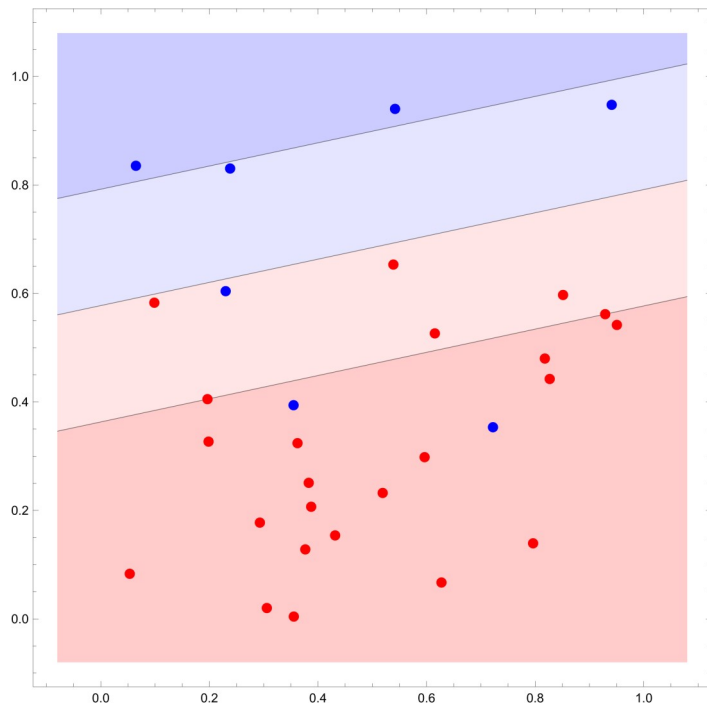
$$\mathbf{w}_2^* = 3\alpha = 3$$

Frage 14

Richtig

Erreichte Punkte 4,00 von 4,00

Consider a C-SVM classifier trained with a linear kernel on a simple 2D data set (red dots = positive samples, blue dots = negative samples) as depicted below. Blue areas are classified as negative (light blue = inside margin) and red areas are classified as positive (light red = inside margin).



Which of the following statements are correct? Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ If the Lagrange multiplier is 0 for a data point, then the corresponding slack variable must be 0. ✓
- ☐ Just considering the red data points: Exactly 4 red points are misclassified.
- ☐ Just considering the blue data points: There are exactly 3 blue support vectors.
- ☒ If the slack variable is larger than 0 for a data point, then also the corresponding Lagrange multiplier must be larger than 0. ✓
- ☐ Considering all data points: Exactly 7 data points require a slack variable larger than 0.

Frage 15

Richtig

Erreichte Punkte 4,00 von 4,00

Consider multi-class classification with Support Vector Machines. Let there be M classes, i.e., $y_i \in \{1, 2, \dots, M\}$. Which of the following statements are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Training pairwise classifiers is usually less time costly due to smaller training sets. ✓
- ☐ b. In the one-versus-one approach, the final classification is done by choosing the class which produces the largest value of all classifiers.
- ☐ c. In the one-versus-all approach, the final classification is done by majority vote.
- ☐ d. In the one-versus-all approach, there are $M-1$ classifiers.
- ☒ e. In the one-versus-one approach, there are $M(M-1)/2$ classifiers. ✓
- ☒ f. In SVM-based multi-class classification, a multi-class problem is divided into many binary classification tasks. ✓

Frage 16

Teilweise richtig

Erreichte Punkte 3,00 von 4,00

Which statements about decision trees are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. The number of leaf nodes in a tree cannot be larger than 2^S with S the number of splits. ✗
- ☒ b. A split maximizes the information gain if the produced subsets are completely homogenous. ✓
- ☒ c. Decision tree learning algorithms recursively split the training set into subsets. ✓
- ☐ d. The maximal Gini impurity of a data set with M classes is $1/M$.
- ☒ e. The maximal entropy of a data set with M classes is $\ln M$. ✓
- ☐ f. Information gain is a common splitting criterion for regression.
- ☐ g. Decision trees can be applied to categorical, but not to numerical features.
- ☒ h. The decision tree complexity is adjusted via the stopping criterion and via pruning. ✓

Frage 17

Richtig

Erreichte Punkte 4,00 von 4,00

Consider a data set $\mathbf{Z}$ with 3 different class labels $y \in \{1, 2, 3\}$. There are $|\mathbf{Z}| = 10^6$ data samples in the set. The probabilities p_k that a label in the data set belongs to class k are given as follows:

$$p_1 = 0.1$$

$$p_2 = 0.1$$

$$p_3 = 0.8$$

Assume that a split s perfectly splits the data set into three homogeneous subsets $\mathbf{Z}_{s,t}$ such that every $\mathbf{Z}_{s,t}$ contains all data samples with label t ($t \in \{1, 2, 3\}$) and no other samples.

Let I_G denote the Gini impurity. Calculate the Gini purity gain $g_G(\mathbf{Z}, s) = I_G(\mathbf{Z}) - \sum_{t=1}^3 \frac{|\mathbf{Z}_{s,t}|}{|\mathbf{Z}|} I_G(\mathbf{Z}_{s,t})$.

You can use a pocket calculator. Give the answer as a real number between 0 and 1, rounded to two decimal places. Do not give the answer in %. Use "." for the decimal point. Number format example: 0.67

Antwort: **Frage 18**

Vollständig

Nicht bewertet

This text field is optional and will not influence the automatic grading. Here, you can write down subresults or your thoughts on the specific tasks. It may be used to your advantage at the end of the semester as mentioned in the exam info file.

$$I_G(\mathbf{Z}) = 0.1 \cdot 0.9 + 0.1 \cdot 0.9 + 0.8 \cdot 0.2 = 0.34$$

$$I_G(\mathbf{Z}_{s,1}) = 0 = I_G(\mathbf{Z}_{s,2}) = I_G(\mathbf{Z}_{s,3})$$

$$g(\mathbf{Z}, s) = 0.34$$